

Lily (Shi-Tao) Zhang

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Professional Summary

Master of Science in Biostatistics with a Ph.D. in Biochemistry and Molecular Biology and over six years of biomedical research experience. Skilled in applying statistical methods to clinical and population health data; developing reproducible analytical workflows; evaluating model assumptions; and communicating results to multidisciplinary research teams. Proficient in R, SAS, and Python, with experience in regression modeling, survival analysis, causal inference, and large-scale healthcare datasets.

Technical Skills

Programming: R, SAS, Python

Statistical Methods: Generalized Linear Models, Survival Analysis, Causal Inference, Propensity Score Methods, Longitudinal and Cross-sectional Data Analysis

Data Management and Reporting: Data Cleaning, Data Wrangling, Missing Data Handling, Reproducible Workflows, Quarto **Healthcare Data:** NHANES, HCUP, BRFSS, HRS

Experience

Biostatistics Intern

May 2026–July 2026

Case Western Reserve University, Cleveland, OH

Inflammatory Extracellular Vesicles Promote Immunosenesence and Frailty Risk in People Aging with HIV Infection

- Analyzed longitudinal plasma and extracellular vesicle biomarker data using R.
- Applied generalized linear models, time-to-event methods to identify biomarkers associated with frailty outcome and progression.
- Evaluated model assumptions, performed multiple-testing correction, and developed reproducible analysis pipelines, publication-quality figures, and statistical reports.
- Collaborated with PI and biomedical researchers to translate statistical findings into biologically meaningful conclusions.

Postdoctoral Fellow

March 2022–July 2025

Case Western Reserve University, Cleveland, OH

- Designed and conducted computational and experimental studies of protein structure and function.
- Analyzed complex biological data, created scientific visualizations, and communicated findings through peer-reviewed publications and presentations.
- Collaborated with multidisciplinary researchers and contributed biological domain expertise to study design and interpretation.

Selected Projects

Comparative Effectiveness of Diabetes Medications — NHANES 2017–2020

- Used propensity score matching to compare metformin and sulfonylurea users and reduce treatment-selection bias.
- Built linear and logistic regression models for HbA1c-related outcomes and performed diagnostics and sensitivity analyses.

Prediction of Health Outcomes in Older Adults — HRS 2022

- Processed large-scale survey data, constructed analysis variables, and developed regression models for self-rated health and chronic conditions.
- Evaluated model performance and translated findings into clear summaries for nontechnical audiences.

Survival Analysis of Clinical Outcomes

- Built Cox proportional hazards models in R and SAS, assessed proportional hazards assumptions, and estimated hazard ratios with confidence intervals.
- Generated Kaplan–Meier curves and interpreted time-to-event results for biomedical research questions.

Education

M.S., Biostatistics, Case Western Reserve University, Cleveland, OH

Aug 2026

Ph.D., Biochemistry and Molecular Biology, Jilin University, China

Dec 2021

B.S., Life Science, Jilin University, China

June 2015

Publications and Certification

Zhang S. et al. *Frontiers in Chemistry*, 2021; Zhang S. et al. *Molecules*, 2021

Certification: SAS 9.4 Programming Fundamentals